

PersonalSenseAI

A Privacy-Preserving Personal Behaviour Assistant for On-Device Digital Phenotyping, Workload Reflection, and Human-Centred AI

Muhammad Muneeb

PhD Candidate in Computational Biology

School of Chemistry and Molecular Biosciences, The University of Queensland, Australia

Proposed Postdoctoral Fellowship / Grant Concept Note

Potential host group: Yet to be decided

Email	m.muneeb@uq.edu.au
Alternate email	muneeksiddique007@gmail.com
Phone	[Insert phone number]
ORCID	0000-0002-6506-4430
Google Scholar	Muhammad Muneeb
GitHub	github.com/MuhammadMuneeb007

Project summary

Digital devices, computers, smartphones and wearables capture rich signals about how people work, rest, move, focus and recover. However, most personal analytics tools are either too shallow, too invasive, or too opaque. They often provide simple screen-time statistics, but do not explain behavioural patterns; or they collect sensitive data without sufficient transparency, user control or research-grade validation. At the same time, research in digital phenotyping and human-centred AI shows that passive and active signals can reveal meaningful patterns in workload, stress, sleep, attention, activity and wellbeing when collected ethically and interpreted cautiously.

I propose to develop **PersonalSenseAI**, a privacy-preserving personal behaviour assistant that learns an individual's behavioural rhythms from consent-based, on-device data streams. The system will collect only participant-approved signals, extract privacy-preserving features locally, learn personalised baselines, detect deviations from the user's own patterns, and generate reflective summaries such as: *"Your focus periods were shorter than your usual baseline this week; this coincided with reduced sleep duration and higher self-reported fatigue."*

The system will **not** be a medical diagnostic tool, workplace surveillance system, or mental-health classifier. It will be a research framework for studying whether local, consent-based digital behaviour models can help people understand their work patterns, fatigue, focus, stress and recovery in a transparent and privacy-preserving manner.

Central idea: Build an on-device personal AI assistant that converts participant-controlled digital traces into interpretable behavioural summaries, while avoiding raw surveillance, clinical overclaiming and black-box inference.

Scientific problem and gap

Modern knowledge work and study are increasingly mediated through screens, mobile devices, online meetings, calendars and digital tasks. People often experience fatigue, distraction, stress or reduced productivity, but they rarely have access to a trustworthy and personalised explanation of their behavioural patterns. Existing tools usually show simple statistics, such as screen time, steps or sleep duration. These summaries are useful but limited because they do not connect multiple signals across time, context and self-report.

Research-grade digital phenotyping has shown that smartphones and wearables can support the study of stress, sleep, activity, sociability and wellbeing. However, several gaps remain:

- many systems rely on centralised data collection, creating privacy and governance concerns;
- models trained across users may generalise poorly because behaviour and physiology vary strongly between individuals;
- raw signals such as screen content, audio, video, keystrokes and precise location can be highly sensitive;
- most systems focus on prediction accuracy rather than explanation, user control and reflective value;
- there is limited work on combining computer-use behaviour, optional wearable data, self-report, calendar/task context and personalised summarisation in a single on-device framework.

The gap is therefore:

People need a privacy-preserving and personalised assistant that can learn from their own consented behaviour data, identify interpretable patterns, and generate useful self-reflection reports without becoming surveillance or clinical diagnosis.

Overall aim

The overall aim is to develop and validate **PersonalSenseAI**, an on-device personal behaviour assistant for consent-based digital phenotyping, workload reflection and privacy-preserving behavioural modelling.

The main research question is:

Can a personal AI assistant learn interpretable behavioural patterns from consent-based digital traces and self-reports while minimising privacy risk through on-device feature extraction, data minimisation and user-controlled reporting?

Specific goals

1. **Design a consent-first data collection framework** that clearly defines what data are collected, what data are never collected, and what participants can turn on or off.
2. **Develop on-device feature extraction modules** for computer interaction patterns, optional smartphone/wearable signals, self-reports, calendar/task context and environmental context.
3. **Build personalised baseline models** that learn each participant's normal rhythms rather than relying only on population-level labels.
4. **Develop interpretable behavioural summaries** that explain changes in focus, workload, fatigue, routine, recovery and digital fragmentation.
5. **Validate the system in staged studies** beginning with public datasets, followed by a small feasibility study and a larger consented longitudinal pilot.
6. **Evaluate privacy, usability and trust** using participant feedback, ablation studies, missing-data analysis and data-minimisation experiments.

Where the information will come from

The project will use a staged data strategy. The first stage will use public datasets to build and test modelling methods. The second and third stages will collect data directly from consenting adult participants through a controlled research protocol approved by the relevant ethics committee.

Stage 0: Public datasets for method development

Before collecting any new participant data, the project will test models on public digital phenotyping and wearable datasets. These datasets are useful for developing baseline models, stress/fatigue classification pipelines, missing-data handling and personalisation methods.

Dataset type	Example sources	Use in the project
Smartphone sensing and self-report	StudentLife-style datasets	Develop pipelines for phone-derived behavioural features, self-report alignment, stress/mood/fatigue prediction and longitudinal modelling.
Wearable stress and affect	WESAD-style datasets	Develop physiological and motion feature extraction, stress/non-stress classification and multimodal fusion models.
Workplace sensing and wellbeing	Tesseract-style information-worker datasets	Study how multimodal sensing relates to affect, stress, work patterns and wellbeing in knowledge-worker settings.
Open human-activity datasets	Activity-recognition and sleep/activity datasets	Pre-train or benchmark movement, routine and activity segmentation modules.

Stage 1: Small feasibility study

A small feasibility study will recruit approximately 10–20 consenting adult participants. The aim will be to test whether the software can run reliably, whether participants understand the data collection process, and whether the generated summaries are useful and not intrusive.

Participants may be recruited from university students, researchers, PhD students, postdoctoral researchers or knowledge workers, depending on the host institution's ethics approval and recruitment pathways. No minors will be recruited. No employer or supervisor will receive individual participant data.

Stage 2: Longitudinal pilot study

A larger pilot study will recruit approximately 50–100 consenting adult participants for 6–12 weeks of data collection. The exact sample size, recruitment route and duration will be finalised with the host group and ethics committee. The study will collect active self-reports and passive digital features under participant control.

Participant data streams

The most important part of this project is defining exactly what information will come from people. The system will collect **features**, not uncontrolled raw surveillance data. Participants will choose which modules are enabled.

Planned participant data streams, privacy controls and purpose

Data stream	Where it comes from	What will be collected	What will not be collected / purpose
Onboarding profile	Participant questionnaire	Age band, role category, study/work pattern, timezone, typical sleep/work schedule, device types, consent preferences.	No exact home address, no identity documents. Used to personalise baselines and interpret routines.
Self-report check-ins	Short mobile/desktop surveys	Fatigue, perceived stress, mood valence, focus, workload, sleep quality, social energy and optional free-text reflection.	Free text will be optional. Used as ground truth for model training and validation.

Data stream	Where it comes from	What will be collected	What will not be collected / purpose
Computer activity features	Local desktop agent	Active/idle time, application category, window-switch rate, focus-session duration, break frequency, meeting/coding/writing categories, typing rhythm summaries.	No passwords, no typed content, no raw keystrokes, no screen recording. Used to model digital fragmentation and focus rhythms.
Calendar/task context	Optional calendar/task integration	Event duration, meeting density, task category, deadline proximity and work-block structure.	No attendee names or private event text by default. Used to contextualise workload.
Smartphone behaviour	Optional phone module	Screen unlocks, screen-on duration, app category duration, notification count, coarse mobility/routine indicators.	No message content, call content, SMS content or precise GPS by default. Used to infer routine and digital interruption patterns.
Wearable physiology	Optional wearable integration	Heart rate summaries, heart-rate variability summaries, sleep duration, activity, steps and recovery proxies where available.	No medical diagnosis. Used to study workload, recovery and fatigue associations.
Environmental context	Optional device sensors or manual entry	Time of day, day of week, session length, optional noise/light level summaries if available.	No raw audio. Used to contextualise concentration and fatigue patterns.
Optional web-cam/microphone features	Separate explicit consent only	On-device derived features such as face-presence, posture proxy, speaking activity or voice-energy summaries.	No raw video or raw audio stored by default. This module can be excluded entirely if ethics risk is too high.
Reflection notes	Optional participant diary	Participant-written notes about workload, fatigue, study/work events or recovery.	Not required; participant can delete entries. Used to improve interpretability of summaries.

What the system will not do

- It will **not** diagnose depression, anxiety, burnout or any medical condition.
- It will **not** provide clinical treatment recommendations.
- It will **not** be designed for employer surveillance or performance monitoring.
- It will **not** collect raw passwords, private messages, typed text, screenshots, raw audio or raw video by default.
- It will **not** share individual-level participant summaries with supervisors, employers or third parties.
- It will **not** make high-stakes decisions about a person.

How the proposed system will work

Step 1: Consent and participant control

Participants will first see a clear consent dashboard. Each data stream will be explained in plain language. Participants will be able to enable or disable modules separately. The consent process will state what is collected, why it is collected, how it is processed, how long it is kept, who can access it, and how participants can withdraw.

Step 2: On-device data collection

A local desktop/mobile agent will collect approved signals. Where possible, raw signals will remain on the participant's device. The system will convert raw events into aggregate features such as session duration, window-switch frequency, active/idle ratios, sleep duration, heart-rate summaries and self-report labels.

Step 3: Privacy-preserving feature extraction

The system will apply data minimisation. For example, instead of storing the name of a document or webpage, the system may store only a broad category such as *writing*, *coding*, *meeting*, *administration*, *communication* or *break*. Instead of storing keystrokes, it will store aggregate typing rhythm features such as burst length and pause frequency. Instead of storing raw audio, it may store only speaking/non-speaking duration if participants explicitly enable that module.

Step 4: Personal baseline modelling

The system will learn each participant's baseline. The aim is not to compare people against each other, but to compare a participant with their own usual patterns. For example:

- usual start/end times for work sessions;
- usual focus-session duration;
- usual meeting density;
- usual digital interruption rate;
- usual sleep and activity patterns;
- usual self-reported focus, stress and fatigue levels.

Step 5: Behavioural pattern detection

The system will identify interpretable patterns such as:

- increased context switching compared with the participant's baseline;
- shorter focus periods on high-meeting days;
- fatigue reports following reduced sleep or extended late-night sessions;
- improved focus after longer breaks or exercise periods;
- recurring workload peaks before deadlines;
- mismatch between perceived productivity and objective work-session structure.

Step 6: Reflective report generation

The assistant will generate daily, weekly or monthly reports. The reports will be written carefully as reflective summaries rather than medical or psychological conclusions.

Example report wording: "This week, your average focused work block was 34 minutes, compared with your 6-week baseline of 48 minutes. The largest reduction occurred on days with three or more meetings. You also reported higher fatigue on two of those days. This may suggest that meeting density is associated with shorter focus blocks for you, but this is only a behavioural association and not a diagnosis."

System architecture

Modelling approach

The modelling strategy will be cautious and interpretable. The project will compare simple baselines with more advanced models rather than beginning with a black-box system.

Module	Function
Consent manager	Presents data options, records consent choices, supports withdrawal and module-level opt-out.
Local data collector	Captures approved desktop, smartphone, wearable and self-report data streams.
Feature extractor	Converts raw events into privacy-preserving aggregate behavioural features.
Personal baseline engine	Learns participant-specific norms and detects within-person deviations.
Multimodal fusion model	Combines self-report, computer behaviour, phone/wearable and calendar/task features.
Explanation layer	Identifies which signals contributed to a pattern or summary.
Reflection report generator	Produces daily/weekly summaries, uncertainty statements and trend visualisations.
Privacy guard	Blocks sensitive fields, removes raw content, applies local processing, encryption and data-retention rules.
Research export layer	Exports de-identified aggregate features only when consent and ethics approval allow.

1. **Feature engineering and descriptive baselines:** active time, idle time, session duration, app-category distribution, meeting density, sleep duration, wearable summaries and self-report trends.
2. **Within-person baseline models:** rolling averages, exponentially weighted moving averages, change-point detection and anomaly detection against the participant's own history.
3. **Supervised prediction:** predict self-reported high fatigue, high workload, low focus or poor sleep using participant-approved features.
4. **Personalisation:** compare population-level models with user-specific fine-tuning and retrieval from the participant's own prior history.
5. **Interpretable machine learning:** use feature attribution, ablation and counterfactual summaries to explain why a pattern was detected.
6. **Report generation:** use rule-based templates and constrained language-model summarisation grounded only in approved features and evidence tables.

Validation plan

Validation question	Evaluation approach
Does the system run reliably?	Measure missing data, battery impact, CPU/memory use, failure rate and participant dropout.
Can signals predict self-reports?	Predict high/low fatigue, focus, workload and stress from approved features using cross-validation and leave-one-participant-out evaluation.
Does personalisation help?	Compare global models, participant-specific models and hybrid models.
Are reports useful?	Ask participants to rate clarity, usefulness, accuracy, intrusiveness and trust.
Are reports safe and cautious?	Review generated summaries for overclaiming, diagnosis-like language and inappropriate recommendations.
Which data streams matter?	Perform ablation studies removing computer, phone, wearable, calendar or self-report signals.
Can privacy be improved without losing utility?	Compare model performance using rawer features versus minimised aggregate features.
Does the system generalise?	Evaluate across students, researchers or knowledge workers depending on recruitment.

Ethics, privacy and governance

This project involves human participants and potentially sensitive behavioural data. Ethics and governance will therefore be central to the design, not an afterthought.

Principle	Implementation
Informed consent	Participants will receive plain-language information about every data stream before collection begins.
Data minimisation	The system will collect the minimum features needed for the research question.
Local processing	Raw signals will be processed locally where possible, with only aggregate features exported under consent.
Participant control	Participants can pause collection, disable modules, view their own data and withdraw.
No surveillance use	Individual-level summaries will not be shared with employers, supervisors or third parties.
No clinical claims	Outputs will be framed as behavioural self-reflection and research evidence, not medical diagnosis.
Security	Data will be encrypted at rest and in transit; identifiers will be separated from feature tables.
Transparency	Reports will explain what signals contributed to each summary and include uncertainty statements.

Connection to my existing work

Although this project is distinct from my genomics work, it builds on my broader strengths in AI systems, research software, benchmarking, evidence integration and automated reporting.

Existing experience	Relevance to PersonalSenseAI
AI-assisted literature and evidence extraction	Supports the design of transparent report generation and evidence-grounded summaries.
JARVIS / local research-agent workflows	Provides experience building local agents that call tools, manage outputs, produce logs and generate reports.
Benchmarking frameworks	Supports rigorous evaluation, ablation studies, reproducibility checks and metric-driven validation.
Bioinformatics pipeline development	Provides experience with heterogeneous data integration, identifier harmonisation and quality control.
Teaching in programming, machine learning and digital technologies	Supports development of a user-facing system understandable to students, researchers and knowledge workers.
Digital behaviour monitoring concept	Provides the starting idea for privacy-preserving behavioural summaries and personal reflection reports.

Expected outputs

The project will produce:

1. an open-source local personal behaviour assistant prototype;
2. a consent-first data collection and governance framework;
3. privacy-preserving feature extraction modules for desktop, mobile, wearable and self-report data;
4. personalised baseline models for workload, focus, fatigue and routine patterns;
5. interpretable daily/weekly reflection reports;
6. a de-identified aggregate research dataset if permitted by ethics approval and participant consent;

7. a methods paper on privacy-preserving personal digital phenotyping;
8. a validation paper on personalised modelling of fatigue, workload and focus in students or knowledge workers;
9. documentation, tutorials and reusable analysis notebooks.

Feasibility and risk management

Risk	Mitigation
Privacy concerns	Use opt-in modules, local processing, aggregate features, encryption and participant control.
Ethics complexity	Begin with public datasets, then run a small feasibility study before larger data collection.
Participant dropout	Minimise burden, keep surveys short and provide useful participant-facing summaries.
Sensor noise and missing data	Use robust features, imputation, missingness indicators and within-person baselines.
Overclaiming mental-health inference	Avoid diagnostic language and position outputs as self-reflection and research summaries.
Battery or performance burden	Use lightweight collection intervals and benchmark resource usage.
Model generalisation issues	Compare population, personalised and hybrid models; report uncertainty and limitations.
Misuse for surveillance	Explicitly prohibit employer/supervisor access to individual-level data and design for personal use only.

Fit with potential host groups

This project would fit host groups working in human-computer interaction, ubiquitous computing, digital health, affective computing, privacy-preserving AI, wearable sensing, behavioural data science, mental-health informatics or human-centred machine learning.

The project can be adapted depending on the host:

- **HCI / ubiquitous computing host:** focus on consent, interaction design, usability, participant control and reflective interfaces.
- **Digital health host:** focus on stress, fatigue, sleep, wellbeing and longitudinal behavioural modelling.
- **Privacy / security host:** focus on local computation, data minimisation, privacy-preserving features and governance.
- **Machine-learning host:** focus on personalised modelling, multimodal fusion, missing-data robustness and interpretable AI.

Indicative timetable

Why this project is important

PersonalSenseAI addresses a real problem in modern work and study: people generate behavioural data every day, but they rarely receive useful, transparent and privacy-preserving insight from it. A well-designed personal assistant could help people understand their own patterns without turning their life into surveillance. The scientific contribution is not simply building an app. The contribution is a validated framework for consent-based, on-device, personalised behavioural modelling and reflective AI summaries.

The long-term vision is a system that helps a person ask:

Period	Planned activities
Months 1–3	Finalise host alignment, ethics plan, data streams, consent materials and public-dataset baselines.
Months 4–6	Build local data collector, self-report module, privacy guard and feature extraction layer.
Months 7–9	Develop baseline models using public datasets and simulated/local data streams.
Months 10–12	Run small feasibility study; evaluate usability, missing data, technical reliability and participant burden.
Months 13–15	Refine system; implement personalised baseline models, report generator and explanation layer.
Months 16–18	Launch longitudinal pilot study with consented participants.
Months 19–21	Analyse predictive validity, personalisation, privacy-utility trade-offs and participant feedback.
Months 22–24	Finalise open-source release, manuscripts, documentation, de-identified aggregate outputs where permitted and future grant applications.

How do my work patterns, sleep, interruptions, meetings and self-reported fatigue relate to my focus and wellbeing over time?

PersonalSenseAI will provide a cautious, transparent and privacy-preserving way to answer that question.

Selected references

1. Wang, R. et al. StudentLife: assessing mental health, academic performance and behavioural trends using smartphone sensing. *UbiComp* (2014).
2. Schmidt, P. et al. Introducing WESAD, a multimodal dataset for wearable stress and affect detection. *ICMI* (2018).
3. Hosseini, S. et al. A multimodal sensor dataset for continuous stress detection of nurses in a hospital. *Scientific Data* (2022).
4. Sano, A. et al. The Tesseract Project: large-scale, longitudinal, in-situ, multimodal sensing of information workers. *CHI Extended Abstracts* (2019).
5. Onnela, J.-P. and Rauch, S. L. Harnessing smartphone-based digital phenotyping to enhance behavioural and mental health. *Neuropsychopharmacology* (2016).
6. Torous, J. et al. Digital phenotyping and mobile sensing in mental health research. *World Psychiatry* (2018).
7. National Health and Medical Research Council. *National Statement on Ethical Conduct in Human Research*. Australia (2025).
8. World Health Organization. *Ethics and governance of artificial intelligence for health*. WHO (2021).